

LAB 3: ARCHITECTURE SELECTION & JUSTIFICATION

System: Multi-Vendor Artisan E-Commerce Marketplace

Architecture Selection: Layered Architecture

We chose Layered Architecture for the Multi-Vendor Artisan E-Commerce Marketplace.

ARCHITECTURAL CHOICE:

The system is divided into three layers: Presentation Layer, Business Layer, and Data Layer. The Presentation Layer handles shopper and artisan vendor interactions. The Business Layer manages product catalog, orders, payments, and vendor management. The Data Layer stores marketplace information in the database.

Reason 1:

Layered Architecture provides clear separation of responsibilities. Each layer handles a specific type of functionality, making the marketplace easier to understand and maintain.

Reason 2:

The architecture supports modular development. Services such as Product Catalog, Order Manager, Payment Service, and Vendor Management can communicate through defined interfaces.

SECURITY ADVANTAGE:

The layered structure helps separate user-facing functions from business and data operations. Access to the database can be controlled through the business layer instead of allowing direct access from the users.

PERFORMANCE BENEFIT:

The Product Catalog Service can handle product-related requests separately from order and payment operations. This separation helps organize processing and allows frequently accessed catalog data to be handled efficiently.